

GRRLIB

4.6.1

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Chapter 1

GRRLIB Documentation

Welcome to the GRRLIB documentation. A complete list of functions is available from this [page](#).

1.1 Introduction

GRRLIB is a C/C++ 2D/3D graphics library for Wii application developers. It is essentially a wrapper which presents a friendly interface to the Nintendo GX core.

1.2 Links

Code: <https://github.com/GRRLIB/GRRLIB>

Discussions: <https://github.com/GRRLIB/GRRLIB/discussions>

1.3 Credits

Project Leader : NoNameNo

Documentation : Crayon, BlueChip

Lead Coder : NoNameNo

Support Coders : Crayon, Xane, DragonMinded, BlueChip

Advisors : RedShade, Jespa

1.4 Licence

See the [LICENCE](#) file for licence rights and limitations (MIT).

Chapter 2

LICENCE

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Chapter 3

Changelog

All notable changes to this project will be documented in this file.

Unreleased

- Add `GRRLIB_PrintfSystemFont()` to print text using the ROM font on non-Japanese systems.
- Allow rendering of paletted PNG images.
- Use `SYS_MainLoop()` in all examples.

4.6.1 - 2025-08-02

- Fixed Makefile installation.

4.6.0 - 2025-02-16

- Allow compilation and installation with cmake.
- Added `GRRLIB_LoadTextureTPL()` to load TPL data (Texture Palette Library) into a `GRRLIB_texImg`.
- Added `GRRLIB_CreateEmptyTextureFmt()` to create an empty texture with a given format.

4.5.1 - 2024-03-02

- Changed the libraries installation so it always creates the directory structure before moving the files.

4.5.0 - 2024-02-11

- Fixed compatibility issues with devkitPPC release 39.
- Fixed linking problem in examples with FreeType 2.13.2 (pkgrel 1 and 2).
- Added `GRRLIB_LoadTTFFromFile()` to load a TTF from a file.
- Added `GRRLIB_Ellipse()` to draw an ellipse.
- Changed `GRRLIB_PrintfTTF()` and `GRRLIB_PrintfTTFW()` so they use the alpha channel from the `color` parameter.
- Changed function arguments types in a few functions.
- Fixed documentation for `GRRLIB_Camera3dSettings()`, `GRRLIB_Screen2Texture()` and `GRRLIB_CompoEnd()`.

4.4.1 - 2021-03-05

- Patched widescreen on Wii U.

4.4.0 - 2020-01-12

- Removed libpng, zlib, libfreetype and libjpeg from the project. These libraries should now be installed in devkitPro with pacman.
- GRRLIB and pngu are now installed into the *portlibs* folder instead of the *libogc* folder.
- Removed `uint` declaration. The `u32` type should be used instead.
- The `GRRLIB_CreateEmptyTexture()` function is not inline anymore.

4.3.2 - 2012-08-12

- libpng was updated to version 1.5.12
- zlib was updated to version 1.2.7
- libjpeg was updated to version 8d
- FreeType updated to version 2.4.10
- `GRRLIB_ClampVar8()` was removed from GRRLIB
- Removed warnings from `GRRLIB_LoadBMF()`

4.3.1 - 2010-10-22

- libpng was updated to version 1.4.4
- FreeType updated to version 2.4.3
- GRRLIB is compatible with libogc 1.8.4

4.3.0 - 2010-06-28

- libpng was updated to version 1.4.2
- libjpeg was updated to version 8b
- zlib was updated to version 1.2.5
- FreeType 2.3.12 support (first support)
- 3D primitive drawing: torus, sphere, cube, cylinder and cone
- Light functions (diffuse, specular, spot) + ambient
- Split functions for rotation/translation/scaling 3d object.
- Here is the list a new added samples demo code:
 - 3d_light1 -> Simple Diffuse light sample code
 - 3d_light2 -> Simple Lights and `GRRLIB_ObjectViewInv()` sample
 - 3d_light3 -> A little Specular light sample code
 - 3d_light4 -> Spot Light Sample Code
 - 3D_sample5 -> Simple demo of rotation/translation/scaling 3d object.
 - ttf -> TrueType Font demo

4.2.0 - 2009-12-16

- First support to 3D functions
- `GRRLIB_CompoStart()` and `GRRLIB_CompoEnd()` for real GX compositing with transparency support
- `GRRLIB_Screen2Texture()` is now fully optimized
- USB_Gecko output facilities
- `GRRLIB_Compose()` was deleted since it was not fully using GX
- `GRRLIB_GetColor()` was deleted, the RGBA macro should be used instead
- Lot of new sample code:
 - 3D_CubedTileDemo (How to use dynamic texturing)
 - 3D_sample1 (A simple rotating flat cube)
 - 3D_sample2 (A simple rotating textured cube)
 - 3D_sample3 (A textured cube and compositing)
 - 3D_sample4 (A complex object rotating)
 - basic_drawing (How to use some basic GRRLIB functions)
 - bitmap_fx (Effects ShowRoom)
 - blending (How to use blending mode)
 - compositing (A simple compositing how to)
 - funsin (A gradient sinusoid dancing)
 - particle (A nice particle sample code)
 - template (Use this as a basis for your project)
 - TileDemo (This will show you how to use tiles/tileset and map)
 - unlimited2d (A faky technic for unlimited sprites)
 - unlimited3d (Same as above but with 3D)

4.1.1 - 2009-11-24

- Fully compatible with devkitPro release 18 and 19 (code and examples)
- libpng was updated to version 1.2.40
- Support for MS-Windows Bitmap format uncompressed (1-bit, 4-bit, 8-bit, 24-bit and 32-bit)
- New function called `GRRLIB_DrawPart()` to draw a specific part of a texture
- Extra parameters to `GRRLIB_Screen2Texture()`
- Video is now initialized even without a SD card
- Fixed a problem with scaling images
- Documentation improvement

4.1.0 - 2009-10-05

- Completely new file structure with sub-folders
- Installer for vendor libraries (jpeg, png, pngu)
- Makefile for GRRLIB
- Many (all suitable) functions are now inlined
- Support for the recent changes to libogc
- Alpha compositor function added
- Correct use of pointers (no more struct passing)
- Speed improvement, bug fixing and more...

4.0.0 - 2009-03-05

- Changed color format to RGBA for ALL GRRLib functions to fit to `GXColor` format and use `GX_↔Color1u32`
- `GRRLIB_LoadTexture()` now auto detect PNG or JPEG
- GRRLib introduce a new texture structure (easier to handle texture width, height, etc ...)
- Add `GRRLIB_InitTileSet()` to initialize a tile set
- `GRRLIB_DrawImg()` recoded for simpler use
- `GRRLIB_DrawTile()` recoded for simpler use
- `InitVideo()` and `GRRLIB_Start()` merge into `GRRLIB_Init()`.
- Add `GRRLIB_PtInRect()`, `GRRLIB_RectInRect()` and `GRRLIB_RectOnRect()` to detect hot zone
- `GRRLIB_GetPixelFromtexImg()` and `GRRLIB_SetPixelTotexImg()` to directly read/write in texture
- `GRRLIB_CreateEmptyTexture()` and `GRRLIB_FlushTex()`
- New Bitmap FX
- Add `GRRLIB_Exit()` to free the memory allocated by GRRLIB

Chapter 4

Topic Index

4.1 Topics

Here is a list of all topics with brief descriptions:

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Chapter 5

Data Structure Index

5.1 Data Structures

Here are the data structures with brief descriptions:

GRRLIB_bytemapChar	
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Structure to hold the TTF information	59
GRRLIB_texImg	
Structure to hold the texture information	59

Chapter 6

File Index

6.1 File List

Here is a list of all documented files with brief descriptions:

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GRRLIB__inline.h	GRRLIB inline function prototypes	63
GRRLIB__lib.h	GRRLIB library function prototypes	64

Chapter 7

Topic Documentation

7.1 Everything in GRRLIB

This is the complete list of functions, structures, defines, typedefs, enumerations and variables you may want to used to make your homebrew with GRRLIB.

Data Structures

- struct [GRRLIB_drawSettings](#)
Structure to hold the current drawing settings.
- struct [GRRLIB_texImg](#)
Structure to hold the texture information.
- struct [GRRLIB_bytemapChar](#)
Structure to hold the bytemap character information.
- struct [GRRLIB_bytemapFont](#)
Structure to hold the bytemap font information.
- struct [GRRLIB_Font](#)
Structure to hold the TTF information.

Macros

- #define **GRRLIB_VER_STRING** "4.6.1"
Version information for GRRLIB.
- #define [R](#)(c)
Extract red component of colour.
- #define [G](#)(c)
Extract green component of colour.
- #define [B](#)(c)
Extract blue component of colour.
- #define [A](#)(c)
Extract alpha component of colour.
- #define [RGBA](#)(r, g, b, a)
Build an RGB pixel from components.
- #define **GRRLIB_BLEND_NONE** ([GRRLIB_BLEND_ALPHA](#))

- *Alias for `GRRLIB_BLEND_ALPHA`.*
- `#define GRRLIB_BLEND_LIGHT (GRRLIB\_BLEND\_ADD)`
- *Alias for `GRRLIB_BLEND_ADD`.*
- `#define GRRLIB_BLEND_SHADE (GRRLIB\_BLEND\_MULTI)`
- *Alias for `GRRLIB_BLEND_MULTI`.*
- `#define GRR_EXTERN extern`
- `#define GRR_INIT(v)`
- `#define GRR_INITS(...)`
- `#define INLINE static inline`

Typedefs

- typedef enum [GRRLIB_blendMode](#) **GRRLIB_blendMode**
GRRLIB Blending Modes.
- typedef struct GRRLIB_drawSettings **GRRLIB_drawSettings**
Structure to hold the current drawing settings.
- typedef struct GRRLIB_texImg **GRRLIB_texImg**
Structure to hold the texture information.
- typedef struct GRRLIB_bytemapChar **GRRLIB_bytemapChar**
Structure to hold the bytemap character information.
- typedef struct GRRLIB_bytemapFont **GRRLIB_bytemapFont**
Structure to hold the bytemap font information.
- typedef struct [GRRLIB_Font](#) **GRRLIB_ttfFont**
Structure to hold the TTF information.

Enumerations

- enum [GRRLIB_blendMode](#) {
[GRRLIB_BLEND_ALPHA](#) = 0 , [GRRLIB_BLEND_ADD](#) = 1 , [GRRLIB_BLEND_SCREEN](#) = 2 ,
[GRRLIB_BLEND_MULTI](#) = 3 ,
[GRRLIB_BLEND_INV](#) = 4 }
GRRLIB Blending Modes.

Functions

- `GRR_EXTERN void *xfb[2] GRR_INITS (NULL, NULL)`
- `GRR_EXTERN u32 fb GRR_INIT (0)`
- `INLINE void GRRLIB_ClipReset (void)`
Reset the clipping to normal.
- `INLINE void GRRLIB\_ClipDrawing (const u32 x, const u32 y, const u32 width, const u32 height)`
Clip the drawing area to an rectangle.
- `INLINE bool GRRLIB\_PtInRect (const int hotx, const int hoty, const int hotw, const int hoth, const int wpadx, const int wpady)`
Determine whether the specified point lies within the specified rectangle.
- `INLINE bool GRRLIB\_RectInRect (const int rect1x, const int rect1y, const int rect1w, const int rect1h, const int rect2x, const int rect2y, const int rect2w, const int rect2h)`
Determine whether a specified rectangle lies within another rectangle.
- `INLINE bool GRRLIB\_RectOnRect (const int rect1x, const int rect1y, const int rect1w, const int rect1h, const int rect2x, const int rect2y, const int rect2w, const int rect2h)`
Determine whether a part of a specified rectangle lies on another rectangle.

- `INLINE void GRRLIB_NPlot (const guVector v[], const u32 color[], const u16 n)`
Draw an array of points.
- `INLINE void GRRLIB_NGone (const guVector v[], const u32 color[], const u16 n)`
Draw a polygon.
- `INLINE void GRRLIB_NGoneFilled (const guVector v[], const u32 color[], const u16 n)`
Draw a filled polygon.
- `INLINE void GRRLIB_GXEngine (const guVector v[], const u32 color[], const u16 n, const u8 fmt)`
Draws a vector.
- `INLINE void GRRLIB_FillScreen (const u32 color)`
Clear screen with a specific color.
- `INLINE void GRRLIB_Plot (const f32 x, const f32 y, const u32 color)`
Draw a dot.
- `INLINE void GRRLIB_Line (const f32 x1, const f32 y1, const f32 x2, const f32 y2, const u32 color)`
Draw a line.
- `INLINE void GRRLIB_Rectangle (const f32 x, const f32 y, const f32 width, const f32 height, const u32 color, const bool filled)`
Draw a rectangle.
- `INLINE void GRRLIB_SetHandle (GRRLIB_texImg *tex, const int x, const int y)`
Set a texture's X and Y handles.
- `INLINE void GRRLIB_SetMidHandle (GRRLIB_texImg *tex, const bool enabled)`
Center a texture's handles.
- `INLINE u32 GRRLIB_GetPixelFromtexImg (const int x, const int y, const GRRLIB_texImg *tex)`
Return the color value of a pixel from a GRRLIB_texImg.
- `INLINE void GRRLIB_SetPixelTotexImg (const int x, const int y, GRRLIB_texImg *tex, const u32 color)`
Set the color value of a pixel to a GRRLIB_texImg.
- `INLINE u32 GRRLIB_GetPixelFromFB (int x, int y)`
Reads a pixel directly from the FrontBuffer.
- `INLINE void GRRLIB_SetPixelToFB (int x, int y, u32 pokeColor)`
Writes a pixel directly from the FrontBuffer.
- `INLINE void GRRLIB_SetBlend (const GRRLIB_blendMode blendmode)`
Set a blending mode.
- `INLINE GRRLIB_blendMode GRRLIB_GetBlend (void)`
Get the current blending mode.
- `INLINE void GRRLIB_SetAntiAliasing (const bool aa)`
Turn anti-aliasing on/off.
- `INLINE bool GRRLIB_GetAntiAliasing (void)`
Get current anti-aliasing setting.
- `INLINE void GRRLIB_ClearTex (GRRLIB_texImg *tex)`
Clear a texture to transparent black.
- `INLINE void GRRLIB_FlushTex (GRRLIB_texImg *tex)`
Write the contents of a texture in the data cache down to main memory.
- `INLINE void GRRLIB_FreeTexture (GRRLIB_texImg *tex)`
Free memory allocated for texture.
- `GRRLIB_bytemapFont * GRRLIB_LoadBMF (const u8 my_bmf[])`
Load a ByteMap font structure from a buffer.
- `void GRRLIB_FreeBMF (GRRLIB_bytemapFont *bmf)`
Free memory allocated by ByteMap fonts.
- `void GRRLIB_InitTileSet (GRRLIB_texImg *tex, const u32 tilew, const u32 tileh, const u32 tilestart)`
Initialize a tile set.
- `void GRRLIB_BMFX_FlipH (const GRRLIB_texImg *texsrc, GRRLIB_texImg *texdest)`
Flip texture horizontal.

- void [GRRLIB_BMFX_FlipV](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest)
Flip texture vertical.
- void [GRRLIB_BMFX_Grayscale](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest)
Change a texture to gray scale.
- void [GRRLIB_BMFX_Sepia](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest)
Change a texture to sepia (old photo style).
- void [GRRLIB_BMFX_Invert](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest)
Invert colors of the texture.
- void [GRRLIB_BMFX_Blur](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)
A texture effect (Blur).
- void [GRRLIB_BMFX_Scatter](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)
A texture effect (Scatter).
- void [GRRLIB_BMFX_Pixelate](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)
A texture effect (Pixelate).
- int [GRRLIB_Init](#) (void)
Initialize GRRLIB.
- void [GRRLIB_Exit](#) (void)
Call this before exiting your application.
- void [GRRLIB_Ellipse](#) (const f32 x, const f32 y, const f32 radiusX, const f32 radiusY, const u32 color, const u8 filled)
Draw an ellipse.
- void [GRRLIB_Circle](#) (const f32 x, const f32 y, const f32 radius, const u32 color, const u8 filled)
Draw a circle.
- int [GRRLIB_LoadFile](#) (const char *filename, u8 **data)
Load a file to memory.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureFromFile](#) (const char *filename)
Load a texture from a file.
- [GRRLIB_ttfFont](#) * [GRRLIB_LoadTTFFromFile](#) (const char *filename)
Load a TTF from a file.
- bool [GRRLIB_ScrShot](#) (const char *filename)
Make a PNG screenshot.
- void [GRRLIB_Printf](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const u32 color, const f32 zoom, const char *text,...)
Print formatted output.
- void [GRRLIB_PrintBMF](#) (const f32 xpos, const f32 ypos, const [GRRLIB_bytemapFont](#) *bmf, const char *text,...)
Print formatted output with a ByteMap font.
- void [GRRLIB_DrawImg](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color)
Draw a texture.
- void [GRRLIB_DrawImgQuad](#) (const guVector pos[4], [GRRLIB_texImg](#) *tex, const u32 color)
Draw a textured quad.
- void [GRRLIB_DrawTile](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color, const int frame)
Draw a tile.
- void [GRRLIB_DrawPart](#) (const f32 xpos, const f32 ypos, const f32 partx, const f32 party, const f32 partw, const f32 parth, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color)
Draw a part of a texture.
- void [GRRLIB_DrawTileQuad](#) (const guVector pos[4], [GRRLIB_texImg](#) *tex, const u32 color, const int frame)
Draw a tile in a quad.
- void [GRRLIB_Render](#) (void)

- Call this function after drawing.*

 - void [GRRLIB_Screen2Texture](#) (u16 posx, u16 posy, [GRRLIB_texImg](#) *tex, bool clear)

Make a snapshot of the screen in a texture WITHOUT ALPHA LAYER.
- void [GRRLIB_CompStart](#) (void)

Start GX compositing process.
- void [GRRLIB_CompEnd](#) (u16 posx, u16 posy, [GRRLIB_texImg](#) *tex)

End GX compositing process (Make a snapshot of the screen in a texture WITH ALPHA LAYER).
- [GRRLIB_texImg](#) * [GRRLIB_CreateEmptyTexture](#) (const u32 width, const u32 height)

Create an empty texture in GX_TF_RGBA8 format.
- [GRRLIB_texImg](#) * [GRRLIB_CreateEmptyTextureFmt](#) (const u32 width, const u32 height, const u32 format)

Create an empty texture with a given format.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTexture](#) (const u8 *my_img)

Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTexturePNG](#) (const u8 *my_png)

Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureJPG](#) (const u8 *my_jpg)

Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureJPGEx](#) (const u8 *my_jpg, const u32 my_size)

Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureBMP](#) (const u8 *my_bmp)

Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureTPL](#) (const u8 *my_tpl, u32 id)

Set TPL data to a [GRRLIB_texImg](#) structure.
- bool [GRRLIB_GeckoInit](#) (void)

Initialize USB Gecko.
- void [GRRLIB_GeckoPrintf](#) (const char *text,...)

Print Gecko.
- void [GRRLIB_SetBackgroundColour](#) (u8 r, u8 g, u8 b, u8 a)

Set the background parameter when screen is cleared.
- void [GRRLIB_Camera3dSettings](#) (f32 posx, f32 posy, f32 posz, f32 upx, f32 upy, f32 upz, f32 lookx, f32 looky, f32 lookz)

Set the camera parameter (contributed by chris_c aka DaShAmAn).
- void [GRRLIB_3dMode](#) (f32 minDist, f32 maxDist, f32 fov, bool texturemode, bool normalmode)

Set up the position matrix (contributed by chris_c aka DaShAmAn).
- void **GRRLIB_2dMode** (void)

Go back to 2D mode (contributed by chris_c aka DaShAmAn).
- void **GRRLIB_ObjectViewBegin** (void)

Init the object matrix to draw object.
- void [GRRLIB_ObjectViewScale](#) (f32 scalx, f32 scaly, f32 scalz)

Scale the object matrix to draw object.
- void [GRRLIB_ObjectViewRotate](#) (f32 angx, f32 angy, f32 angz)

Rotate the object matrix to draw object.
- void [GRRLIB_ObjectViewTrans](#) (f32 posx, f32 posy, f32 posz)

Translate the object matrix to draw object.
- void **GRRLIB_ObjectViewEnd** (void)

Concat the object and the view matrix and calculate the inverse normal matrix.
- void [GRRLIB_ObjectView](#) (f32 posx, f32 posy, f32 posz, f32 angx, f32 angy, f32 angz, f32 scalx, f32 scaly, f32 scalz)

Set the view matrix to draw object (in this order scale, rotate AND trans).
- void [GRRLIB_ObjectViewInv](#) (f32 posx, f32 posy, f32 posz, f32 angx, f32 angy, f32 angz, f32 scalx, f32 scaly, f32 scalz)

- Set the view matrix to draw object (in this order scale, trans AND rotate).*

 - void `GRRLIB_SetTexture` (`GRRLIB_texImg *tex`, bool rep)
- Set the texture to an object (contributed by chris_c aka DaShAmAn).*

 - void `GRRLIB_DrawTorus` (f32 r, f32 R, int nsides, int rings, bool filled, u32 col)
- Draw a torus (with normal).*

 - void `GRRLIB_DrawSphere` (f32 r, int lats, int longs, bool filled, u32 col)
- Draw a sphere (with normal).*

 - void `GRRLIB_DrawCube` (f32 size, bool filled, u32 col)
- Draw a cube (with normal).*

 - void `GRRLIB_DrawCylinder` (f32 r, f32 h, u16 d, bool filled, u32 col)
- Draw a cylinder (with normal).*

 - void `GRRLIB_DrawCone` (f32 r, f32 h, u16 d, bool filled, u32 col)
- Draw a cone (with normal).*

 - void `GRRLIB_DrawTessPanel` (f32 w, f32 wstep, f32 h, f32 hstep, bool filled, u32 col)
- Draw a Tesselated panel (with normal).*

 - void `GRRLIB_SetLightAmbient` (u32 ambientcolor)
- Set ambient color.*

 - void `GRRLIB_SetLightDiff` (u8 num, guVector pos, f32 distattn, f32 brightness, u32 lightcolor)
- Set diffuse light parameters.*

 - void `GRRLIB_SetLightSpec` (u8 num, guVector dir, f32 shininess, u32 lightcolor, u32 speccolor)
- Set specular light parameters.*

 - void `GRRLIB_SetLightSpot` (u8 num, guVector pos, guVector lookat, f32 angAttn0, f32 angAttn1, f32 angAttn2, f32 distAttn0, f32 distAttn1, f32 distAttn2, u32 lightcolor)
- Set Spot light parameters.*

 - void `GRRLIB_SetLightOff` (void)
- Set all lights off, like at init.*

 - `GRRLIB_ttfFont * GRRLIB_LoadTTF` (const u8 *file_base, s32 file_size)
- Load a TTF from a buffer.*

 - void `GRRLIB_FreeTTF` (`GRRLIB_ttfFont *myFont`)
- Free memory allocated by TTF fonts.*

 - void `GRRLIB_PrintfTTF` (int x, int y, `GRRLIB_ttfFont *myFont`, const char *string, unsigned int fontSize, const u32 color)
- Print function for TTF font.*

 - void `GRRLIB_PrintfTTFW` (int x, int y, `GRRLIB_ttfFont *myFont`, const wchar_t *string, unsigned int fontSize, const u32 color)
- Print function for TTF font.*

 - u32 `GRRLIB_WidthTTF` (`GRRLIB_ttfFont *myFont`, const char *, unsigned int)
- Get the width of a text in pixel.*

 - u32 `GRRLIB_WidthTTFW` (`GRRLIB_ttfFont *myFont`, const wchar_t *, unsigned int)
- Get the width of a text in pixel.*

 - void `GRRLIB_PrintfSystemFont` (int x, int y, const char *string, unsigned int fontSize, const u32 color)
- Print text using the system font.*

Variables

- `GRR_EXTERN GXRModeObj * rmode`
- Video mode.*

7.1.1 Detailed Description

This is the complete list of functions, structures, defines, typedefs, enumerations and variables you may want to use to make your homebrew with GRRLIB.

You simply need to include [grrlib.h](#) in your project to have access to all of these.

7.1.2 Macro Definition Documentation

7.1.2.1 A

```
#define A(  
    c)
```

Value:

```
((c) & 0xFF)
```

Extract alpha component of colour.

7.1.2.2 B

```
#define B(  
    c)
```

Value:

```
((c) >> 8) & 0xFF)
```

Extract blue component of colour.

7.1.2.3 G

```
#define G(  
    c)
```

Value:

```
((c) >> 16) & 0xFF)
```

Extract green component of colour.

7.1.2.4 R

```
#define R(  
    c)
```

Value:

```
((c) >> 24) & 0xFF)
```

Extract red component of colour.

Parameters

<i>r</i>	Red component.
<i>g</i>	Green component.
<i>b</i>	Blue component.
<i>a</i>	Alpha component.

7.1.3 Enumeration Type Documentation**7.1.3.1 GRRLIB_blendMode**

enum [GRRLIB_blendMode](#)

GRRLIB Blending Modes.

Enumerator

GRRLIB_BLEND_ALPHA	Alpha Blending.
GRRLIB_BLEND_ADD	Additive Blending.
GRRLIB_BLEND_SCREEN	Alpha Light Blending.
GRRLIB_BLEND_MULTI	Multiply Blending.
GRRLIB_BLEND_INV	Invert Color Blending.

7.1.4 Function Documentation**7.1.4.1 GRRLIB_3dMode()**

```
void GRRLIB_3dMode (
    f32 minDist,
    f32 maxDist,
    f32 fov,
    bool texturemode,
    bool normalmode)
```

Set up the position matrix (contributed by [chris_c](#) aka DaShAmAn).

Parameters

<i>minDist</i>	Minimal distance for the camera.
<i>maxDist</i>	Maximal distance for the camera.
<i>fov</i>	Field of view for the camera.
<i>texturemode</i>	False, GX won't need texture coordinate, True, GX will need texture coordinate.
<i>normalmode</i>	False, GX won't need normal coordinate, True, GX will need normal coordinate.

7.1.4.2 GRRLIB_BMFX_Blur()

```
void GRRLIB_BMFX_Blur (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest,
    const u32 factor)
```

A texture effect (Blur).

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.
<i>factor</i>	The blur factor.

7.1.4.3 GRRLIB_BMFX_FlipH()

```
void GRRLIB_BMFX_FlipH (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest)
```

Flip texture horizontal.

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.

7.1.4.4 GRRLIB_BMFX_FlipV()

```
void GRRLIB_BMFX_FlipV (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest)
```

Flip texture vertical.

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.

7.1.4.5 GRRLIB_BMFX_Grayscale()

```
void GRRLIB_BMFX_Grayscale (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest)
```

Change a texture to gray scale.

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture grayscaled destination.

7.1.4.6 GRRLIB_BMFX_Invert()

```
void GRRLIB_BMFX_Invert (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest)
```

Invert colors of the texture.

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.

7.1.4.7 GRRLIB_BMFX_Pixelate()

```
void GRRLIB_BMFX_Pixelate (
    const GRRLIB_texImg * texsrc,
    GRRLIB_texImg * texdest,
    const u32 factor)
```

A texture effect (Pixelate).

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.
<i>factor</i>	The factor level of the effect.

7.1.4.8 GRRLIB_BMFX_Scatter()

```
void GRRLIB_BMFX_Scatter (  
    const GRRLIB_texImg * texsrc,  
    GRRLIB_texImg * texdest,  
    const u32 factor)
```

A texture effect (Scatter).

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.
<i>factor</i>	The factor level of the effect.

7.1.4.9 GRRLIB_BMFX_Sepia()

```
void GRRLIB_BMFX_Sepia (  
    const GRRLIB_texImg * texsrc,  
    GRRLIB_texImg * texdest)
```

Change a texture to sepia (old photo style).

See also

[GRRLIB_FlushTex](#)

Parameters

<i>texsrc</i>	The texture source.
<i>texdest</i>	The texture destination.

Author

elisherer

7.1.4.10 GRRLIB_Camera3dSettings()

```
void GRRLIB_Camera3dSettings (
    f32 posx,
    f32 posy,
    f32 posz,
    f32 upx,
    f32 upy,
    f32 upz,
    f32 lookx,
    f32 looky,
    f32 lookz)
```

Set the camera parameter (contributed by `chris_c` aka DaShAmAn).

Parameters

<i>posx</i>	x position of the camera.
<i>posy</i>	y position of the camera.
<i>posz</i>	z position of the camera.
<i>upx</i>	x up position of the camera.
<i>upy</i>	y up position of the camera.
<i>upz</i>	z up position of the camera.
<i>lookx</i>	x position of the target.
<i>looky</i>	y position of the target.
<i>lookz</i>	z position of the target.

7.1.4.11 GRRLIB_Circle()

```
void GRRLIB_Circle (
    const f32 x,
    const f32 y,
    const f32 radius,
    const u32 color,
    const u8 filled)
```

Draw a circle.

Parameters

<i>x</i>	Specifies the x-coordinate of the circle.
<i>y</i>	Specifies the y-coordinate of the circle.
<i>radius</i>	The radius of the circle.
<i>color</i>	The color of the circle in RGBA format.
<i>filled</i>	Set to <code>true</code> to fill the circle.

7.1.4.12 GRRLIB_ClearTex()

```
INLINE void GRRLIB_ClearTex (
    GRRLIB_texImg * tex)
```

Clear a texture to transparent black.

Parameters

<i>tex</i>	Texture to clear.
------------	-------------------

7.1.4.13 GRRLIB_ClipDrawing()

```

INLINE void GRRLIB_ClipDrawing (
    const u32 x,
    const u32 y,
    const u32 width,
    const u32 height)

```

Clip the drawing area to an rectangle.

Parameters

<i>x</i>	The x-coordinate of the rectangle.
<i>y</i>	The y-coordinate of the rectangle.
<i>width</i>	The width of the rectangle.
<i>height</i>	The height of the rectangle.

7.1.4.14 GRRLIB_CompEnd()

```

void GRRLIB_CompEnd (
    u16 posx,
    u16 posy,
    GRRLIB_texImg * tex)

```

End GX compositing process (Make a snapshot of the screen in a texture WITH ALPHA LAYER).

EFB is cleared after this function.

See also

[GRRLIB_CompStart](#)

Parameters

<i>posx</i>	Top left corner of the grabbed part.
<i>posy</i>	Top left corner of the grabbed part.
<i>tex</i>	A pointer to a texture representing the screen.

7.1.4.15 GRRLIB_CompStart()

```

void GRRLIB_CompStart (
    void )

```

Start GX compositing process.

See also

[GRRLIB_CompEnd](#)

7.1.4.16 GRRLIB_CreateEmptyTexture()

```
GRRLIB_texImg * GRRLIB_CreateEmptyTexture (
    const u32 width,
    const u32 height)
```

Create an empty texture in GX_TF_RGBA8 format.

Parameters

<i>width</i>	Width of the new texture to create.
<i>height</i>	Height of the new texture to create.

Returns

A [GRRLIB_texImg](#) structure newly created.

7.1.4.17 GRRLIB_CreateEmptyTextureFmt()

```
GRRLIB_texImg * GRRLIB_CreateEmptyTextureFmt (
    const u32 width,
    const u32 height,
    const u32 format)
```

Create an empty texture with a given format.

Parameters

<i>width</i>	Width of the new texture to create.
<i>height</i>	Height of the new texture to create.
<i>format</i>	Format of the new texture to create.

Returns

A [GRRLIB_texImg](#) structure newly created.

7.1.4.18 GRRLIB_DrawCone()

```
void GRRLIB_DrawCone (
    f32 r,
    f32 h,
    u16 d,
    bool filled,
    u32 col)
```

Draw a cone (with normal).

Parameters

<i>r</i>	Radius of the cone.
<i>h</i>	High of the cone.
<i>d</i>	Density of slice.
<i>filled</i>	Wired or not.
<i>col</i>	Color of the cone.

7.1.4.19 GRRLIB_DrawCube()

```
void GRRLIB_DrawCube (
    f32 size,
    bool filled,
    u32 col)
```

Draw a cube (with normal).

Parameters

<i>size</i>	Size of the cube edge.
<i>filled</i>	Wired or not.
<i>col</i>	Color of the cube.

7.1.4.20 GRRLIB_DrawCylinder()

```
void GRRLIB_DrawCylinder (
    f32 r,
    f32 h,
    u16 d,
    bool filled,
    u32 col)
```

Draw a cylinder (with normal).

Parameters

<i>r</i>	Radius of the cylinder.
<i>h</i>	High of the cylinder.
<i>d</i>	Density of slice.
<i>filled</i>	Wired or not.
<i>col</i>	Color of the cylinder.

7.1.4.21 GRRLIB_DrawImg()

```
void GRRLIB_DrawImg (
    const f32 xpos,
    const f32 ypos,
    const GRRLIB_texImg * tex,
    const f32 degrees,
    const f32 scaleX,
    const f32 scaleY,
    const u32 color)
```

Draw a texture.

Parameters

<i>xpos</i>	Specifies the x-coordinate of the upper-left corner.
<i>ypos</i>	Specifies the y-coordinate of the upper-left corner.
<i>tex</i>	The texture to draw.
<i>degrees</i>	Angle of rotation.
<i>scaleX</i>	Specifies the x-coordinate scale. -1 could be used for flipping the texture horizontally.
<i>scaleY</i>	Specifies the y-coordinate scale. -1 could be used for flipping the texture vertically.
<i>color</i>	Color in RGBA format.

7.1.4.22 GRRLIB_DrawImgQuad()

```
void GRRLIB_DrawImgQuad (
    const guVector pos[4],
    GRRLIB_texImg * tex,
    const u32 color)
```

Draw a textured quad.

Parameters

<i>pos</i>	Vector array of the 4 points.
<i>tex</i>	The texture to draw.
<i>color</i>	Color in RGBA format.

7.1.4.23 GRRLIB_DrawPart()

```
void GRRLIB_DrawPart (
    const f32 xpos,
    const f32 ypos,
    const f32 partx,
    const f32 party,
    const f32 partw,
    const f32 parth,
    const GRRLIB_texImg * tex,
    const f32 degrees,
    const f32 scaleX,
    const f32 scaleY,
    const u32 color)
```

Draw a part of a texture.

Parameters

<i>xpos</i>	Specifies the x-coordinate of the upper-left corner.
<i>ypos</i>	Specifies the y-coordinate of the upper-left corner.
<i>partx</i>	Specifies the x-coordinate of the upper-left corner in the texture.
<i>party</i>	Specifies the y-coordinate of the upper-left corner in the texture.
<i>partw</i>	Specifies the width in the texture.
<i>parth</i>	Specifies the height in the texture.
<i>tex</i>	The texture containing the tile to draw.
<i>degrees</i>	Angle of rotation.
<i>scaleX</i>	Specifies the x-coordinate scale. -1 could be used for flipping the texture horizontally.
<i>scaleY</i>	Specifies the y-coordinate scale. -1 could be used for flipping the texture vertically.
<i>color</i>	Color in RGBA format.

7.1.4.24 GRRLIB_DrawSphere()

```
void GRRLIB_DrawSphere (
    f32 r,
    int lats,
    int longs,
    bool filled,
    u32 col)
```

Draw a sphere (with normal).

Parameters

<i>r</i>	Radius of the sphere.
<i>lats</i>	Number of latitudes.
<i>longs</i>	Number of longitudes.
<i>filled</i>	Wired or not.
<i>col</i>	Color of the sphere.

7.1.4.25 GRRLIB_DrawTessPanel()

```
void GRRLIB_DrawTessPanel (
    f32 w,
    f32 wstep,
    f32 h,
    f32 hstep,
    bool filled,
    u32 col)
```

Draw a Tesselated panel (with normal).

Parameters

<i>w</i>	Width of the panel.
<i>wstep</i>	Size of width slices.
<i>h</i>	Height of the panel.
<i>hstep</i>	Size the de height slices.
<i>filled</i>	Wired or not.
<i>col</i>	Color in RGBA format.

7.1.4.26 GRRLIB_DrawTile()

```
void GRRLIB_DrawTile (
    const f32 xpos,
    const f32 ypos,
    const GRRLIB_texImg * tex,
    const f32 degrees,
    const f32 scaleX,
    const f32 scaleY,
    const u32 color,
    const int frame)
```

Draw a tile.

Parameters

<i>xpos</i>	Specifies the x-coordinate of the upper-left corner.
<i>ypos</i>	Specifies the y-coordinate of the upper-left corner.
<i>tex</i>	The texture containing the tile to draw.
<i>degrees</i>	Angle of rotation.
<i>scaleX</i>	Specifies the x-coordinate scale. -1 could be used for flipping the texture horizontally.
<i>scaleY</i>	Specifies the y-coordinate scale. -1 could be used for flipping the texture vertically.
<i>color</i>	Color in RGBA format.
<i>frame</i>	Specifies the frame to draw.

7.1.4.27 GRRLIB_DrawTileQuad()

```
void GRRLIB_DrawTileQuad (
    const guVector pos[4],
    GRRLIB_texImg * tex,
    const u32 color,
    const int frame)
```

Draw a tile in a quad.

Parameters

<i>pos</i>	Vector array of the 4 points.
------------	-------------------------------

<i>tex</i>	The texture to draw.
<i>color</i>	Color in RGBA format.
<i>frame</i>	Specifies the frame to draw.

7.1.4.28 GRRLIB_DrawTorus()

```
void GRRLIB_DrawTorus (  
    f32 r,  
    f32 R,  
    int nsides,  
    int rings,  
    bool filled,  
    u32 col)
```

Draw a torus (with normal).

Parameters

<i>r</i>	Radius of the ring.
<i>R</i>	Radius of the torus.
<i>nsides</i>	Number of faces per ring.
<i>rings</i>	Number of rings.
<i>filled</i>	Wired or not.
<i>col</i>	Color of the torus.

7.1.4.29 GRRLIB_Ellipse()

```
void GRRLIB_Ellipse (  
    const f32 x,  
    const f32 y,  
    const f32 radiusX,  
    const f32 radiusY,  
    const u32 color,  
    const u8 filled)
```

Draw an ellipse.

Author

Dark_Link

Parameters

<i>x</i>	Specifies the x-coordinate of the ellipse.
<i>y</i>	Specifies the y-coordinate of the ellipse.
<i>radiusX</i>	The X radius of the ellipse.

<i>radiusY</i>	The Y radius of the ellipse.
<i>color</i>	The color of the ellipse in RGBA format.
<i>filled</i>	Set to <code>true</code> to fill the ellipse.

7.1.4.30 GRRLIB_Exit()

```
void GRRLIB_Exit (
    void )
```

Call this before exiting your application.

Ensure this function is only ever called once and only if the setup function has been called.

7.1.4.31 GRRLIB_FillScreen()

```
INLINE void GRRLIB_FillScreen (
    const u32 color)
```

Clear screen with a specific color.

Parameters

<i>color</i>	The color to use to fill the screen.
--------------	--------------------------------------

7.1.4.32 GRRLIB_FlushTex()

```
INLINE void GRRLIB_FlushTex (
    GRRLIB_texImg * tex)
```

Write the contents of a texture in the data cache down to main memory.

For performance the CPU holds a data cache where modifications are stored before they get written down to main memory.

Parameters

<i>tex</i>	The texture to flush.
------------	-----------------------

7.1.4.33 GRRLIB_FreeBMF()

```
void GRRLIB_FreeBMF (
    GRRLIB_bytemapFont * bmf)
```

Free memory allocated by ByteMap fonts.

If *bmf* is a null pointer, the function does nothing.

Note

This function does not change the value of *bmf* itself, hence it still points to the same (now invalid) location.

Parameters

<i>bmf</i>	A GRRLIB_bytemapFont structure.
------------	---

7.1.4.34 GRRLIB_FreeTexture()

```

INLINE void GRRLIB_FreeTexture (
    GRRLIB\_texImg * tex)

```

Free memory allocated for texture.

If *tex* is a null pointer, the function does nothing.

Note

This function does not change the value of *tex* itself, hence it still points to the same (now invalid) location.

Parameters

<i>tex</i>	A GRRLIB_texImg structure.
------------	--

7.1.4.35 GRRLIB_FreeTTF()

```

void GRRLIB_FreeTTF (
    GRRLIB\_ttfFont * myFont)

```

Free memory allocated by TTF fonts.

If *myFont* is a null pointer, the function does nothing.

Note

This function does not change the value of *myFont* itself, hence it still points to the same (now invalid) location.

Parameters

<i>myFont</i>	A TTF.
---------------	--------

7.1.4.36 GRRLIB_GeckoInit()

```

bool GRRLIB_GeckoInit (
    void )

```

Initialize USB Gecko.

Returns

Returns **true** if everything worked, **false** if problems occurred.

7.1.4.37 GRRLIB_GeckoPrintf()

Parameters

<i>text</i>	Text to print.
...	Optional arguments.

7.1.4.38 GRRLIB_GetAntiAliasing()

```

INLINE bool GRRLIB_GetAntiAliasing (
    void )

```

Get current anti-aliasing setting.

Returns

Returns `true` if anti-aliasing is enabled.

7.1.4.39 GRRLIB_GetBlend()

```

INLINE GRRLIB_blendMode GRRLIB_GetBlend (
    void )

```

Get the current blending mode.

Returns

The current blending mode.

7.1.4.40 GRRLIB_GetPixelFromFB()

```

INLINE u32 GRRLIB_GetPixelFromFB (
    int x,
    int y)

```

Reads a pixel directly from the FrontBuffer.

Parameters

<i>x</i>	The x-coordinate within the FB.
<i>y</i>	The y-coordinate within the FB.

Returns

The color of a pixel in RGBA format.

7.1.4.41 GRRLIB_GetPixelFromtexImg()

```

INLINE u32 GRRLIB_GetPixelFromtexImg (
    const int x,
    const int y,
    const GRRLIB_texImg * tex)

```

Parameters

<i>x</i>	Specifies the x-coordinate of the pixel in the texture.
<i>y</i>	Specifies the y-coordinate of the pixel in the texture.
<i>tex</i>	The texture to get the color from.

Returns

The color of a pixel in RGBA format.

7.1.4.42 GRRLIB_GXEngine()

```
INLINE void GRRLIB_GXEngine (
    const guVector v[],
    const u32 color[],
    const u16 n,
    const u8 fmt)
```

Draws a vector.

Parameters

<i>v</i>	The vector to draw.
<i>color</i>	The color of the vector in RGBA format.
<i>n</i>	Number of points in the vector. The maximum is 65536.
<i>fmt</i>	Type of primitive.

7.1.4.43 GRRLIB_Init()

```
int GRRLIB_Init (
    void )
```

Initialize GRRLIB.

Call this once at the beginning your code.

Returns

A integer representing a code:

- 0 : The operation completed successfully.
- -1 : Not enough memory is available to initialize GRRLIB.
- -2 : Failed to add the fat device driver to the devoptab.
- -3 : Failed to initialize the ttf font engine.
- -4 : Failed to initialize the system font engine.

See also

[GRRLIB_Exit](#)

7.1.4.44 GRRLIB_InitTileSet()

```
void GRRLIB_InitTileSet (
    GRRLIB_texImg * tex,
    const u32 tilew,
    const u32 tileh,
    const u32 tilestart)
```

Initialize a tile set.

Parameters

<i>tex</i>	The texture to initialize.
<i>tilew</i>	Width of the tile.
<i>tileh</i>	Height of the tile.
<i>tilestart</i>	Offset for starting position (Used in fonts).

7.1.4.45 GRRLIB_Line()

```
INLINE void GRRLIB_Line (
    const f32 x1,
    const f32 y1,
    const f32 x2,
    const f32 y2,
    const u32 color)
```

Draw a line.

Parameters

<i>x1</i>	Starting point for line for the x coordinate.
<i>y1</i>	Starting point for line for the y coordinate.
<i>x2</i>	Ending point for line for the x coordinate.
<i>y2</i>	Ending point for line for the x coordinate.
<i>color</i>	Line color in RGBA format.

Author

JESPA

7.1.4.46 GRRLIB_LoadBMF()

```
GRRLIB_bytemapFont * GRRLIB_LoadBMF (
    const u8 my_bmf[])
```

Load a ByteMap font structure from a buffer.

File format version 1.1 is used, more information could be found at <https://bmf.php5.cz/?page=format>

Parameters

<i>my_bmf</i>	The ByteMap font buffer to load.
---------------	----------------------------------

Returns

A [GRRLIB_bytemapFont](#) structure filled with BMF information.

See also

[GRRLIB_FreeBMF](#)

7.1.4.47 GRRLIB_LoadFile()

```
int GRRLIB_LoadFile (
    const char * filename,
    u8 ** data)
```

Load a file to memory.

Parameters

<i>filename</i>	Name of the file to be loaded.
<i>data</i>	Pointer-to-your-pointer. Ie. { u8 *data; GRRLIB_LoadFile("file", &data); }. It is your responsibility to free the memory allocated by this function.

Returns

A integer representing a code:

- 0 : EmptyFile.
- -1 : FileNotFound.
- -2 : OutOfMemory.
- -3 : FileReadError.
- >0 : FileLength.

7.1.4.48 GRRLIB_LoadTexture()

```
GRRLIB_texImg * GRRLIB_LoadTexture (
    const u8 * my_img)
```

Load a texture from a buffer.

Parameters

<i>my_img</i>	The JPEG, PNG or Bitmap buffer to load.
---------------	---

Returns

A [GRRLIB_texImg](#) structure filled with image information.

7.1.4.49 GRRLIB_LoadTextureBMP()

```
GRRLIB_texImg * GRRLIB_LoadTextureBMP (  
    const u8 * my_bmp)
```

Load a texture from a buffer.

It only works for the MS-Windows standard format uncompressed (1-bit, 4-bit, 8-bit, 24-bit and 32-bit).

Parameters

<i>my_bmp</i>	The Bitmap buffer to load.
---------------	----------------------------

Returns

A [GRRLIB_texImg](#) structure filled with image information.

7.1.4.50 GRRLIB_LoadTextureFromFile()

```
GRRLIB_texImg * GRRLIB_LoadTextureFromFile (  
    const char * filename)
```

Load a texture from a file.

Parameters

<i>filename</i>	The JPEG, PNG or Bitmap filename to load.
-----------------	---

Returns

A [GRRLIB_texImg](#) structure filled with image information. If an error occurs NULL will be returned.

7.1.4.51 GRRLIB_LoadTextureJPG()

```
GRRLIB_texImg * GRRLIB_LoadTextureJPG (  
    const u8 * my_jpg)
```

Load a texture from a buffer.

Take care to have the JPG finish with 0xFF 0xD9!

Parameters

<i>my_jpg</i>	The JPEG buffer to load.
---------------	--------------------------

Returns

A [GRRLIB_texImg](#) structure filled with image information.

7.1.4.52 GRRLIB_LoadTextureJPGEEx()

```
GRRLIB_texImg * GRRLIB_LoadTextureJPGEEx (  
    const u8 * my_jpg,  
    const u32 my_size)
```

Load a texture from a buffer.

Author

DrTwox

Parameters

<i>my_jpg</i>	The JPEG buffer to load.
<i>my_size</i>	Size of the JPEG buffer to load.

Returns

A [GRRLIB_texImg](#) structure filled with image information.

7.1.4.53 GRRLIB_LoadTexturePNG()

```
GRRLIB_texImg * GRRLIB_LoadTexturePNG (  
    const u8 * my_png)
```

Load a texture from a buffer.

Parameters

<i>my_png</i>	the PNG buffer to load.
---------------	-------------------------

Returns

A [GRRLIB_texImg](#) structure filled with image information. If image size is not correct, the texture will be completely transparent.

7.1.4.54 GRRLIB_LoadTextureTPL()

```
GRRLIB_texImg * GRRLIB_LoadTextureTPL (  
    const u8 * my_tpl,  
    u32 id)
```

Set TPL data to a [GRRLIB_texImg](#) structure.

Parameters

<i>my_tpl</i>	The TPL buffer to set.
<i>id</i>	ID of the TPL buffer to set.

Returns

A [GRRLIB_texImg](#) structure filled with image information.

7.1.4.55 GRRLIB_LoadTTF()

```
GRRLIB_ttFont * GRRLIB_LoadTTF (
    const u8 * file_base,
    s32 file_size)
```

Load a TTF from a buffer.

Parameters

<i>file_base</i>	Buffer with TTF data. You must not deallocate the memory before calling GRRLIB_FreeTTF .
<i>file_size</i>	Size of the TTF buffer.

Returns

A handle to a given TTF font object or NULL if it fails to load the font.

See also

[GRRLIB_FreeTTF](#)

7.1.4.56 GRRLIB_LoadTTFFromFile()

```
GRRLIB_ttFont * GRRLIB_LoadTTFFromFile (
    const char * filename)
```

Load a TTF from a file.

Parameters

<i>filename</i>	The TTF filename to load.
-----------------	---------------------------

Returns

A [GRRLIB_ttFont](#) structure filled with font information. If an error occurs NULL will be returned.

7.1.4.57 GRRLIB_NGone()

```
INLINE void GRRLIB_NGone (
    const guVector v[],
```

Parameters

<i>v</i>	The vector containing the coordinates of the polygon.
<i>color</i>	The color of the filled polygon in RGBA format.
<i>n</i>	Number of points in the vector. The maximum is 65536.

7.1.4.58 GRRLIB_NGoneFilled()

```

INLINE void GRRLIB_NGoneFilled (
    const guVector v[],
    const u32 color[],
    const u16 n)

```

Draw a filled polygon.

Parameters

<i>v</i>	The vector containing the coordinates of the polygon.
<i>color</i>	The color of the filled polygon in RGBA format.
<i>n</i>	Number of points in the vector. The maximum is 65536.

7.1.4.59 GRRLIB_NPlot()

```

INLINE void GRRLIB_NPlot (
    const guVector v[],
    const u32 color[],
    const u16 n)

```

Draw an array of points.

Parameters

<i>v</i>	Array containing the points.
<i>color</i>	The color of the points in RGBA format.
<i>n</i>	Number of points in the vector array. The maximum is 65536.

7.1.4.60 GRRLIB_ObjectView()

```

void GRRLIB_ObjectView (
    f32 posx,
    f32 posy,
    f32 posz,
    f32 angx,
    f32 angy,
    f32 angz,
    f32 scalx,
    f32 scaly,
    f32 scalz)

```

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Set the view matrix to draw object (in this order scale, rotate AND trans).

Parameters

<i>posx</i>	x position of the object.
<i>posy</i>	y position of the object.
<i>posz</i>	z position of the object.
<i>angx</i>	x rotation angle of the object.
<i>angy</i>	y rotation angle of the object.
<i>angz</i>	z rotation angle of the object.
<i>scalx</i>	x scale of the object.
<i>scaly</i>	y scale of the object.
<i>scalz</i>	z scale of the object.

7.1.4.61 GRRLIB_ObjectViewInv()

```
void GRRLIB_ObjectViewInv (
    f32 posx,
    f32 posy,
    f32 posz,
    f32 angx,
    f32 angy,
    f32 angz,
    f32 scalx,
    f32 scaly,
    f32 scalz)
```

Set the view matrix to draw object (in this order scale, trans AND rotate).

Parameters

<i>posx</i>	x position of the object.
<i>posy</i>	y position of the object.
<i>posz</i>	z position of the object.
<i>angx</i>	x rotation angle of the object.
<i>angy</i>	y rotation angle of the object.
<i>angz</i>	z rotation angle of the object.
<i>scalx</i>	x scale of the object.
<i>scaly</i>	y scale of the object.
<i>scalz</i>	z scale of the object.

7.1.4.62 GRRLIB_ObjectViewRotate()

```
void GRRLIB_ObjectViewRotate (
    f32 angx,
    f32 angy,
    f32 angz)
```

Rotate the object matrix to draw object.

Parameters

<i>angx</i>	x rotation angle of the object.
<i>angy</i>	y rotation angle of the object.
<i>angz</i>	z rotation angle of the object.

7.1.4.63 GRRLIB_ObjectViewScale()

```
void GRRLIB_ObjectViewScale (
    f32 scalx,
    f32 scaly,
    f32 scalz)
```

Scale the object matrix to draw object.

Parameters

<i>scalx</i>	x scale of the object.
<i>scaly</i>	y scale of the object.
<i>scalz</i>	z scale of the object.

7.1.4.64 GRRLIB_ObjectViewTrans()

```
void GRRLIB_ObjectViewTrans (
    f32 posx,
    f32 posy,
    f32 posz)
```

Translate the object matrix to draw object.

Parameters

<i>posx</i>	x position of the object.
<i>posy</i>	y position of the object.
<i>posz</i>	z position of the object.

7.1.4.65 GRRLIB_Plot()

```
INLINE void GRRLIB_Plot (
    const f32 x,
    const f32 y,
    const u32 color)
```

Draw a dot.

Parameters

<i>x</i>	Specifies the x-coordinate of the dot.
<i>y</i>	Specifies the y-coordinate of the dot.
<i>color</i>	The color of the dot in RGBA format.

Author

Jespa

7.1.4.66 GRRLIB_PrintBMF()

```
void GRRLIB_PrintBMF (
    const f32 xpos,
    const f32 ypos,
    const GRRLIB_byteMapFont * bmf,
    const char * text,
    ...)
```

Print formatted output with a ByteMap font.

This function could be slow, it should be used with GRRLIB_CompoStart and GRRLIB_CompoEnd.

Parameters

<i>xpos</i>	Specifies the x-coordinate of the upper-left corner of the text.
<i>ypos</i>	Specifies the y-coordinate of the upper-left corner of the text.
<i>bmf</i>	The ByteMap font to use.
<i>text</i>	Text to draw.
...	Optional arguments.

7.1.4.67 GRRLIB_Printf()

```
void GRRLIB_Printf (
    const f32 xpos,
    const f32 ypos,
    const GRRLIB_texImg * tex,
    const u32 color,
    const f32 zoom,
    const char * text,
    ...)
```

Print formatted output.

Parameters

<i>xpos</i>	Specifies the x-coordinate of the upper-left corner of the text.
-------------	--

<i>ypos</i>	Specifies the y-coordinate of the upper-left corner of the text.
<i>tex</i>	The texture containing the character set.
<i>color</i>	Text color in RGBA format. The alpha channel is used to change the opacity of the text.
<i>zoom</i>	This is a factor by which the text size will be increase or decrease.
<i>text</i>	Text to draw.
...	Optional arguments.

7.1.4.68 GRRLIB_PrintfSystemFont()

```
void GRRLIB_PrintfSystemFont (
    int x,
    int y,
    const char * string,
    unsigned int fontSize,
    const u32 color)
```

Print text using the system font.

Parameters

<i>x</i>	Specifies the x-coordinate of the upper-left corner of the text.
<i>y</i>	Specifies the y-coordinate of the upper-left corner of the text.
<i>string</i>	Text to draw.
<i>fontSize</i>	Height of the text in pixels.
<i>color</i>	Text color in RGBA format.

7.1.4.69 GRRLIB_PrintfTTF()

```
void GRRLIB_PrintfTTF (
    int x,
    int y,
    GRRLIB_ttfFont * myFont,
    const char * string,
    unsigned int fontSize,
    const u32 color)
```

Print function for TTF font.

Parameters

<i>x</i>	Specifies the x-coordinate of the upper-left corner of the text.
<i>y</i>	Specifies the y-coordinate of the upper-left corner of the text.
<i>myFont</i>	A TTF.
<i>string</i>	Text to draw.
<i>fontSize</i>	Size of the font.
<i>color</i>	Text color in RGBA format.

7.1.4.70 GRRLIB_PrintfTTFW()

```
void GRRLIB_PrintfTTFW (
    int x,
    int y,
    GRRLIB_ttfFont * myFont,
    const wchar_t * utf32,
    unsigned int fontSize,
    const u32 color)
```

Print function for TTF font.

Author

wplaat and DrTwox

Parameters

<i>x</i>	Specifies the x-coordinate of the upper-left corner of the text.
<i>y</i>	Specifies the y-coordinate of the upper-left corner of the text.
<i>myFont</i>	A TTF.
<i>utf32</i>	Text to draw.
<i>fontSize</i>	Size of the font.
<i>color</i>	Text color in RGBA format.

7.1.4.71 GRRLIB_PtInRect()

```
INLINE bool GRRLIB_PtInRect (
    const int hotx,
    const int hoty,
    const int hotw,
    const int hoth,
    const int wpadx,
    const int wpady)
```

Determine whether the specified point lies within the specified rectangle.

Parameters

<i>hotx</i>	Specifies the x-coordinate of the upper-left corner of the rectangle.
<i>hoty</i>	Specifies the y-coordinate of the upper-left corner of the rectangle.
<i>hotw</i>	The width of the rectangle.
<i>hoth</i>	The height of the rectangle.
<i>wpadx</i>	Specifies the x-coordinate of the point.
<i>wpady</i>	Specifies the y-coordinate of the point.

Returns

If the specified point lies within the rectangle, the return value is `true` otherwise it's `false`.

7.1.4.72 GRRLIB_Rectangle()

```

INLINE void GRRLIB_Rectangle (
    const f32 x,
    const f32 y,
    const f32 width,
    const f32 height,
    const u32 color,
    const bool filled)

```

Draw a rectangle.

Parameters

<i>x</i>	Specifies the x-coordinate of the upper-left corner of the rectangle.
<i>y</i>	Specifies the y-coordinate of the upper-left corner of the rectangle.
<i>width</i>	The width of the rectangle.
<i>height</i>	The height of the rectangle.
<i>color</i>	The color of the rectangle in RGBA format.
<i>filled</i>	Set to <code>true</code> to fill the rectangle.

7.1.4.73 GRRLIB_RectInRect()

```

INLINE bool GRRLIB_RectInRect (
    const int rect1x,
    const int rect1y,
    const int rect1w,
    const int rect1h,
    const int rect2x,
    const int rect2y,
    const int rect2w,
    const int rect2h)

```

Determine whether a specified rectangle lies within another rectangle.

Parameters

<i>rect1x</i>	Specifies the x-coordinate of the upper-left corner of the rectangle.
<i>rect1y</i>	Specifies the y-coordinate of the upper-left corner of the rectangle.
<i>rect1w</i>	Specifies the width of the rectangle.
<i>rect1h</i>	Specifies the height of the rectangle.
<i>rect2x</i>	Specifies the x-coordinate of the upper-left corner of the rectangle.
<i>rect2y</i>	Specifies the y-coordinate of the upper-left corner of the rectangle.
<i>rect2w</i>	Specifies the width of the rectangle.
<i>rect2h</i>	Specifies the height of the rectangle.

Returns

If the specified rectangle lies within the other rectangle, the return value is `true` otherwise it's `false`.

7.1.4.74 GRRLIB_RectOnRect()

```

INLINE bool GRRLIB_RectOnRect (
    const int rect1x,
    const int rect1y,
    const int rect1w,
    const int rect1h,
    const int rect2x,
    const int rect2y,
    const int rect2w,
    const int rect2h)

```

Determine whether a part of a specified rectangle lies on another rectangle.

Parameters

<i>rect1x</i>	Specifies the x-coordinate of the upper-left corner of the first rectangle.
<i>rect1y</i>	Specifies the y-coordinate of the upper-left corner of the first rectangle.
<i>rect1w</i>	Specifies the width of the first rectangle.
<i>rect1h</i>	Specifies the height of the first rectangle.
<i>rect2x</i>	Specifies the x-coordinate of the upper-left corner of the second rectangle.
<i>rect2y</i>	Specifies the y-coordinate of the upper-left corner of the second rectangle.
<i>rect2w</i>	Specifies the width of the second rectangle.
<i>rect2h</i>	Specifies the height of the second rectangle.

Returns

If the specified rectangle lies on the other rectangle, the return value is `true` otherwise it's `false`.

7.1.4.75 GRRLIB_Screen2Texture()

```

void GRRLIB_Screen2Texture (
    u16 posx,
    u16 posy,
    GRRLIB_texImg * tex,
    bool clear)

```

Make a snapshot of the screen in a texture WITHOUT ALPHA LAYER.

Parameters

<i>posx</i>	Top left corner of the grabbed part.
<i>posy</i>	Top left corner of the grabbed part.
<i>tex</i>	A pointer to a texture representing the screen.
<i>clear</i>	When this flag is set to <code>true</code> , the screen is cleared after copy.

7.1.4.76 GRRLIB_ScrShot()

```
bool GRRLIB_ScrShot (
    const char * filename)
```

Make a PNG screenshot.

It should be called after drawing stuff on the screen, but before [GRRLIB_Render\(\)](#). libfat is required to use the function.

Parameters

<i>filename</i>	Name of the file to write.
-----------------	----------------------------

Returns

Returns `true` if everything worked, `false` if problems occurred.

7.1.4.77 GRRLIB_SetAntiAliasing()

```
INLINE void GRRLIB_SetAntiAliasing (
    const bool aa)
```

Turn anti-aliasing on/off.

Parameters

<i>aa</i>	Set to <code>true</code> to enable anti-aliasing (Default: Enabled).
-----------	--

7.1.4.78 GRRLIB_SetBackgroundColour()

```
void GRRLIB_SetBackgroundColour (
    u8 r,
    u8 g,
    u8 b,
    u8 a)
```

Set the background parameter when screen is cleared.

Parameters

<i>r</i>	Red component.
<i>g</i>	Green component.
<i>b</i>	Blue component.
<i>a</i>	Alpha component.

7.1.4.79 GRRLIB_SetBlend()

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```
INLINE void GRRLIB_SetBlend (
    const GRRLIB_blendMode blendmode)
```

Parameters

<i>blendmode</i>	The blending mode to use (Default: <code>GRRLIB_BLEND_ALPHA</code>).
------------------	---

7.1.4.80 GRRLIB_SetHandle()

```

INLINE void GRRLIB_SetHandle (
    GRRLIB_texImg * tex,
    const int x,
    const int y)

```

Set a texture's X and Y handles.

For example, it could be used for the rotation of a texture.

Parameters

<i>tex</i>	The texture to set the handle on.
<i>x</i>	The x-coordinate of the handle.
<i>y</i>	The y-coordinate of the handle.

7.1.4.81 GRRLIB_SetLightAmbient()

```

void GRRLIB_SetLightAmbient (
    u32 ambientcolor)

```

Set ambient color.

When no diffuse light is shining on a object, the color is equal to ambient color.

Parameters

<i>ambientcolor</i>	Ambient color in RGBA format.
---------------------	-------------------------------

7.1.4.82 GRRLIB_SetLightDiff()

```

void GRRLIB_SetLightDiff (
    u8 num,
    guVector pos,
    f32 distattn,
    f32 brightness,
    u32 lightcolor)

```

Set diffuse light parameters.

Parameters

<i>num</i>	Number of the light. It's a number from 0 to 7.
<i>pos</i>	Position of the diffuse light (x/y/z).
<i>distattn</i>	Distance attenuation.
<i>brightness</i>	Brightness of the light. The value should be between 0 and 1.
<i>lightcolor</i>	Color of the light in RGBA format.

7.1.4.83 GRRLIB_SetLightSpec()

```
void GRRLIB_SetLightSpec (
    u8 num,
    guVector dir,
    f32 shininess,
    u32 lightcolor,
    u32 speccolor)
```

Set specular light parameters.

Parameters

<i>num</i>	Number of the light. It's a number from 0 to 7.
<i>dir</i>	Direction of the specular ray (x/y/z).
<i>shininess</i>	Shininess of the specular. (between 4 and 254)
<i>lightcolor</i>	Color of the light in RGBA format.
<i>speccolor</i>	Specular color in RGBA format.

7.1.4.84 GRRLIB_SetLightSpot()

```
void GRRLIB_SetLightSpot (
    u8 num,
    guVector pos,
    guVector lookat,
    f32 angAttn0,
    f32 angAttn1,
    f32 angAttn2,
    f32 distAttn0,
    f32 distAttn1,
    f32 distAttn2,
    u32 lightcolor)
```

Set Spot light parameters.

Parameters

<i>num</i>	Number of the light. It's a number from 0 to 7.
<i>pos</i>	Position of the spot light (x/y/z).

<i>lookat</i>	Where spot light look at (x/y/z).
<i>angAttn0</i>	cone attenuation factor 0.
<i>angAttn1</i>	cone attenuation factor 1.
<i>angAttn2</i>	cone attenuation factor 2.
<i>distAttn0</i>	Distance attenuation factor 0.
<i>distAttn1</i>	Distance attenuation factor 1.
<i>distAttn2</i>	Distance attenuation factor 2.
<i>lightcolor</i>	Color of the light in RGBA format.

7.1.4.85 GRRLIB_SetMidHandle()

```

INLINE void GRRLIB_SetMidHandle (
    GRRLIB_texImg * tex,
    const bool enabled)

```

Center a texture's handles.

For example, it could be used for the rotation of a texture.

Parameters

<i>tex</i>	The texture to center.
<i>enabled</i>	

7.1.4.86 GRRLIB_SetPixelToFB()

```

INLINE void GRRLIB_SetPixelToFB (
    int x,
    int y,
    u32 pokeColor)

```

Writes a pixel directly from the FrontBuffer.

Parameters

<i>x</i>	The x-coordinate within the FB.
<i>y</i>	The y-coordinate within the FB.
<i>pokeColor</i>	The color of the pixel in RGBA format.

7.1.4.87 GRRLIB_SetPixelTotexImg()

```

INLINE void GRRLIB_SetPixelTotexImg (
    const int x,
    const int y,
    GRRLIB_texImg * tex,
    const u32 color)

```

Set the color value of a pixel to a [GRRLIB_texImg](#).

See also

[GRRLIB_FlushTex](#)

Parameters

<i>x</i>	Specifies the x-coordinate of the pixel in the texture.
<i>y</i>	Specifies the y-coordinate of the pixel in the texture.
<i>tex</i>	The texture to set the color to.
<i>color</i>	The color of the pixel in RGBA format.

7.1.4.88 GRRLIB_SetTexture()

```
void GRRLIB_SetTexture (
    GRRLIB_texImg * tex,
    bool rep)
```

Set the texture to an object (contributed by [chris_c](#) aka DaShAmAn).

Parameters

<i>tex</i>	Pointer to an image texture (GRRLIB_texImg format).
<i>rep</i>	Texture Repeat Mode, <code>true</code> will repeat it, <code>false</code> won't.

7.1.4.89 GRRLIB_WidthTTF()

```
u32 GRRLIB_WidthTTF (
    GRRLIB_ttfFont * myFont,
    const char * string,
    unsigned int fontSize)
```

Get the width of a text in pixel.

Parameters

<i>myFont</i>	A TTF.
<i>string</i>	The text to check.
<i>fontSize</i>	The size of the font.

Returns

The width of a text in pixel.

7.1.4.90 GRRLIB_WidthTTFW()

```
u32 GRRLIB_WidthTTFW (
    GRRLIB_ttfFont * myFont,
    const wchar_t * utf32,
    unsigned int fontSize)
```

Get the width of a text in pixel

Parameters

<i>myFont</i>	A TTF.
<i>utf32</i>	The text to check.
<i>fontSize</i>	The size of the font.

Returns

The width of a text in pixel.

Chapter 8

Data Structure Documentation

8.1 GRRLIB_bytemapChar Struct Reference

Structure to hold the bytemap character information.

```
#include <grrlib.h>
```

Data Fields

- **u8 width**
Character width.
- **u8 height**
Character height.
- **s8 relx**
Horizontal offset relative to cursor (-128 to 127).
- **s8 rely**
Vertical offset relative to cursor (-128 to 127).
- **u8 kerning**
Kerning (Horizontal cursor shift after drawing the character).
- **u8 * data**
Character data (uncompressed, 8 bits per pixel).

8.1.1 Detailed Description

Structure to hold the bytemap character information.

The documentation for this struct was generated from the following file:

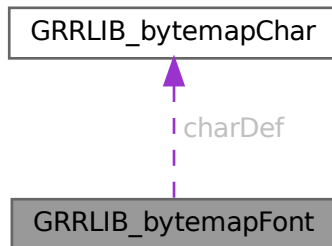
- [grrlib.h](#)

8.2 GRRLIB_bytemapFont Struct Reference

Structure to hold the bytemap font information.

```
#include <grrlib.h>
```

Collaboration diagram for GRRLIB_bytemapFont:



Data Fields

- char * **name**
Font name.
- u32 * **palette**
Font palette.
- u16 **nbChar**
Number of characters in font.
- u8 **version**
Version.
- s8 **tracking**
Tracking (Add-space after each char) (-128 to 127).
- [GRRLIB_bytemapChar](#) **charDef** [256]
Array of bitmap characters.

8.2.1 Detailed Description

Structure to hold the bytemap font information.

The documentation for this struct was generated from the following file:

- [grrlib.h](#)

8.3 GRRLIB_drawSettings Struct Reference

Structure to hold the current drawing settings.

```
#include <grrlib.h>
```

Data Fields

- bool **antialias**
AntiAlias is enabled when set to true.
- [GRRLIB_blendMode](#) **blend**
Blending Mode.
- int **lights**
Active lights.

8.3.1 Detailed Description

Structure to hold the current drawing settings.

The documentation for this struct was generated from the following file:

- [grrlib.h](#)

8.4 GRRLIB_Font Struct Reference

Structure to hold the TTF information.

```
#include <grrlib.h>
```

Data Fields

- void * **face**
A TTF face object.
- bool **kerning**
true whenever a face object contains kerning data that can be accessed with FT_Get_Kerning.

8.4.1 Detailed Description

Structure to hold the TTF information.

The documentation for this struct was generated from the following file:

- [grrlib.h](#)

8.5 GRRLIB_texImg Struct Reference

Structure to hold the texture information.

```
#include <grrlib.h>
```

Data Fields

- **u32 w**
The width of the texture in pixels.
- **u32 h**
The height of the texture in pixels.
- **u32 format**
Texture format.
- **int handlex**
Texture handle x.
- **int handley**
Texture handle y.
- **int offsetx**
Texture offset x.
- **int offsety**
Texture offset y.
- **bool tiledtex**
Texture is tiled if set to true.
- **u32 tilew**
The width of one tile in pixels.
- **u32 tileh**
The height of one tile in pixels.
- **u32 nbtilew**
Number of tiles for the x axis.
- **u32 nbtileh**
Number of tiles for the y axis.
- **u32 tilestart**
Offset to tile starting position.
- **f32 ofnormaltexx**
Offset of normalized texture on x.
- **f32 ofnormaltexy**
Offset of normalized texture on y.
- **void * data**
Pointer to the texture data.

8.5.1 Detailed Description

Structure to hold the texture information.

The documentation for this struct was generated from the following file:

- [grrlib.h](#)

Chapter 9

File Documentation

9.1 grrlib.h File Reference

GRRLIB user include file.

```
#include <gccore.h>
#include "grrlib/GRRLIB__lib.h"
#include "grrlib/GRRLIB__inline.h"
```

Data Structures

- struct [GRRLIB_drawSettings](#)
Structure to hold the current drawing settings.
- struct [GRRLIB_texImg](#)
Structure to hold the texture information.
- struct [GRRLIB_bytemapChar](#)
Structure to hold the bytemap character information.
- struct [GRRLIB_bytemapFont](#)
Structure to hold the bytemap font information.
- struct [GRRLIB_Font](#)
Structure to hold the TTF information.

Macros

- #define [GRRLIB_VER_STRING](#) "4.6.1"
Version information for GRRLIB.
- #define [R\(c\)](#)
Extract red component of colour.
- #define [G\(c\)](#)
Extract green component of colour.
- #define [B\(c\)](#)
Extract blue component of colour.
- #define [A\(c\)](#)
Extract alpha component of colour.

- `#define RGBA(r, g, b, a)`
Build an RGB pixel from components.
- `#define GRRLIB_BLEND_NONE (GRRLIB_BLEND_ALPHA)`
Alias for GRRLIB_BLEND_ALPHA.
- `#define GRRLIB_BLEND_LIGHT (GRRLIB_BLEND_ADD)`
Alias for GRRLIB_BLEND_ADD.
- `#define GRRLIB_BLEND_SHADE (GRRLIB_BLEND_MULTI)`
Alias for GRRLIB_BLEND_MULTI.
- `#define GRR_EXTERN extern`
- `#define GRR_INIT(v)`
- `#define GRR_INITS(...)`
- `#define INLINE static inline`

Typedefs

- `typedef enum GRRLIB_blendMode GRRLIB_blendMode`
GRRLIB Blending Modes.
- `typedef struct GRRLIB_drawSettings GRRLIB_drawSettings`
Structure to hold the current drawing settings.
- `typedef struct GRRLIB_texImg GRRLIB_texImg`
Structure to hold the texture information.
- `typedef struct GRRLIB_bytemapChar GRRLIB_bytemapChar`
Structure to hold the bytemap character information.
- `typedef struct GRRLIB_bytemapFont GRRLIB_bytemapFont`
Structure to hold the bytemap font information.
- `typedef struct GRRLIB_Font GRRLIB_ttfFont`
Structure to hold the TTF information.

Enumerations

- `enum GRRLIB_blendMode {`
`GRRLIB_BLEND_ALPHA = 0 , GRRLIB_BLEND_ADD = 1 , GRRLIB_BLEND_SCREEN = 2 ,`
`GRRLIB_BLEND_MULTI = 3 ,`
`GRRLIB_BLEND_INV = 4 }`
GRRLIB Blending Modes.

Functions

- `GRR_EXTERN void *xfb[2] GRR_INITS (NULL, NULL)`
- `GRR_EXTERN u32 fb GRR_INIT (0)`

Variables

- `GRR_EXTERN GXRModeObj * rmode`
Video mode.

9.1.1 Detailed Description

GRRLIB user include file.

9.2 GRRLIB__inline.h File Reference

GRRLIB inline function prototypes.

```
#include <grrlib/GRRLIB_clipping.h>
#include <grrlib/GRRLIB_collision.h>
#include <grrlib/GRRLIB_fbComplex.h>
#include <grrlib/GRRLIB_fbGX.h>
#include <grrlib/GRRLIB_fbSimple.h>
#include <grrlib/GRRLIB_handle.h>
#include <grrlib/GRRLIB_pixel.h>
#include <grrlib/GRRLIB_settings.h>
#include <grrlib/GRRLIB_texSetup.h>
```

Functions

- INLINE void **GRRLIB_ClipReset** (void)
Reset the clipping to normal.
- INLINE void **GRRLIB_ClipDrawing** (const u32 x, const u32 y, const u32 width, const u32 height)
Clip the drawing area to an rectangle.
- INLINE bool **GRRLIB_PtInRect** (const int hotx, const int hoty, const int hotw, const int hoth, const int wpadx, const int wpady)
Determine whether the specified point lies within the specified rectangle.
- INLINE bool **GRRLIB_RectInRect** (const int rect1x, const int rect1y, const int rect1w, const int rect1h, const int rect2x, const int rect2y, const int rect2w, const int rect2h)
Determine whether a specified rectangle lies within another rectangle.
- INLINE bool **GRRLIB_RectOnRect** (const int rect1x, const int rect1y, const int rect1w, const int rect1h, const int rect2x, const int rect2y, const int rect2w, const int rect2h)
Determine whether a part of a specified rectangle lies on another rectangle.
- INLINE void **GRRLIB_NPlot** (const guVector v[], const u32 color[], const u16 n)
Draw an array of points.
- INLINE void **GRRLIB_NGone** (const guVector v[], const u32 color[], const u16 n)
Draw a polygon.
- INLINE void **GRRLIB_NGoneFilled** (const guVector v[], const u32 color[], const u16 n)
Draw a filled polygon.
- INLINE void **GRRLIB_GXEngine** (const guVector v[], const u32 color[], const u16 n, const u8 fmt)
Draws a vector.
- INLINE void **GRRLIB_FillScreen** (const u32 color)
Clear screen with a specific color.
- INLINE void **GRRLIB_Plot** (const f32 x, const f32 y, const u32 color)
Draw a dot.
- INLINE void **GRRLIB_Line** (const f32 x1, const f32 y1, const f32 x2, const f32 y2, const u32 color)
Draw a line.
- INLINE void **GRRLIB_Rectangle** (const f32 x, const f32 y, const f32 width, const f32 height, const u32 color, const bool filled)
Draw a rectangle.
- INLINE void **GRRLIB_SetHandle** (GRRLIB_texImg *tex, const int x, const int y)
Set a texture's X and Y handles.
- INLINE void **GRRLIB_SetMidHandle** (GRRLIB_texImg *tex, const bool enabled)
Center a texture's handles.
- INLINE u32 **GRRLIB_GetPixelFromtexImg** (const int x, const int y, const GRRLIB_texImg *tex)

- Return the color value of a pixel from a [GRRLIB_texImg](#).*

 - `INLINE void GRRLIB\_SetPixelTotexImg (const int x, const int y, GRRLIB\_texImg *tex, const u32 color)`

Set the color value of a pixel to a [GRRLIB_texImg](#).

 - `INLINE u32 GRRLIB\_GetPixelFromFB (int x, int y)`
- Reads a pixel directly from the FrontBuffer.*
- `INLINE void GRRLIB\_SetPixelToFB (int x, int y, u32 pokeColor)`
- Writes a pixel directly from the FrontBuffer.*
- `INLINE void GRRLIB\_SetBlend (const GRRLIB\_blendMode blendmode)`
- Set a blending mode.*
- `INLINE GRRLIB\_blendMode GRRLIB\_GetBlend (void)`
- Get the current blending mode.*
- `INLINE void GRRLIB\_SetAntiAliasing (const bool aa)`
- Turn anti-aliasing on/off.*
- `INLINE bool GRRLIB\_GetAntiAliasing (void)`
- Get current anti-aliasing setting.*
- `INLINE void GRRLIB\_ClearTex (GRRLIB\_texImg *tex)`
- Clear a texture to transparent black.*
- `INLINE void GRRLIB\_FlushTex (GRRLIB\_texImg *tex)`
- Write the contents of a texture in the data cache down to main memory.*
- `INLINE void GRRLIB\_FreeTexture (GRRLIB\_texImg *tex)`
- Free memory allocated for texture.*

9.2.1 Detailed Description

GRRLIB inline function prototypes.

Do not include [GRRLIB__inline.h](#) directly, include only [GRRLIB.h](#).

9.3 GRRLIB__lib.h File Reference

GRRLIB library function prototypes.

Functions

- `GRRLIB\_bytemapFont * GRRLIB\_LoadBMF (const u8 my_bmf[ ])`
- Load a ByteMap font structure from a buffer.*
- `void GRRLIB\_FreeBMF (GRRLIB\_bytemapFont *bmf)`
- Free memory allocated by ByteMap fonts.*
- `void GRRLIB\_InitTileSet (GRRLIB\_texImg *tex, const u32 tilew, const u32 tileh, const u32 tilestart)`
- Initialize a tile set.*
- `void GRRLIB\_BMFX\_FlipH (const GRRLIB\_texImg *texsrc, GRRLIB\_texImg *texdest)`
- Flip texture horizontal.*
- `void GRRLIB\_BMFX\_FlipV (const GRRLIB\_texImg *texsrc, GRRLIB\_texImg *texdest)`
- Flip texture vertical.*
- `void GRRLIB\_BMFX\_Grayscale (const GRRLIB\_texImg *texsrc, GRRLIB\_texImg *texdest)`
- Change a texture to gray scale.*
- `void GRRLIB\_BMFX\_Sepia (const GRRLIB\_texImg *texsrc, GRRLIB\_texImg *texdest)`

- Change a texture to sepia (old photo style).*

 - void [GRRLIB_BMFX_Invert](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest)

Invert colors of the texture.
- void [GRRLIB_BMFX_Blur](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)

A texture effect (Blur).
- void [GRRLIB_BMFX_Scatter](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)

A texture effect (Scatter).
- void [GRRLIB_BMFX_Pixelate](#) (const [GRRLIB_texImg](#) *texsrc, [GRRLIB_texImg](#) *texdest, const u32 factor)

A texture effect (Pixelate).
- int [GRRLIB_Init](#) (void)

Initialize GRRLIB.
- void [GRRLIB_Exit](#) (void)

Call this before exiting your application.
- void [GRRLIB_Ellipse](#) (const f32 x, const f32 y, const f32 radiusX, const f32 radiusY, const u32 color, const u8 filled)

Draw an ellipse.
- void [GRRLIB_Circle](#) (const f32 x, const f32 y, const f32 radius, const u32 color, const u8 filled)

Draw a circle.
- int [GRRLIB_LoadFile](#) (const char *filename, u8 **data)

Load a file to memory.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureFromFile](#) (const char *filename)

Load a texture from a file.
- [GRRLIB_ttfFont](#) * [GRRLIB_LoadTTFFromFile](#) (const char *filename)

Load a TTF from a file.
- bool [GRRLIB_ScrShot](#) (const char *filename)

Make a PNG screenshot.
- void [GRRLIB_Printf](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const u32 color, const f32 zoom, const char *text,...)

Print formatted output.
- void [GRRLIB_PrintBMF](#) (const f32 xpos, const f32 ypos, const [GRRLIB_bytemapFont](#) *bmf, const char *text,...)

Print formatted output with a ByteMap font.
- void [GRRLIB_DrawImg](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color)

Draw a texture.
- void [GRRLIB_DrawImgQuad](#) (const guVector pos[4], [GRRLIB_texImg](#) *tex, const u32 color)

Draw a textured quad.
- void [GRRLIB_DrawTile](#) (const f32 xpos, const f32 ypos, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color, const int frame)

Draw a tile.
- void [GRRLIB_DrawPart](#) (const f32 xpos, const f32 ypos, const f32 partx, const f32 party, const f32 partw, const f32 parth, const [GRRLIB_texImg](#) *tex, const f32 degrees, const f32 scaleX, const f32 scaleY, const u32 color)

Draw a part of a texture.
- void [GRRLIB_DrawTileQuad](#) (const guVector pos[4], [GRRLIB_texImg](#) *tex, const u32 color, const int frame)

Draw a tile in a quad.
- void [GRRLIB_Render](#) (void)

Call this function after drawing.
- void [GRRLIB_Screen2Texture](#) (u16 posx, u16 posy, [GRRLIB_texImg](#) *tex, bool clear)

Make a snapshot of the screen in a texture WITHOUT ALPHA LAYER.
- void [GRRLIB_CompoStart](#) (void)

Start GX compositing process.

- void [GRRLIB_CompoEnd](#) (u16 posx, u16 posy, [GRRLIB_texImg](#) *tex)
End GX compositing process (Make a snapshot of the screen in a texture WITH ALPHA LAYER).
- [GRRLIB_texImg](#) * [GRRLIB_CreateEmptyTexture](#) (const u32 width, const u32 height)
Create an empty texture in GX_TF_RGBA8 format.
- [GRRLIB_texImg](#) * [GRRLIB_CreateEmptyTextureFmt](#) (const u32 width, const u32 height, const u32 format)
Create an empty texture with a given format.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTexture](#) (const u8 *my_img)
Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTexturePNG](#) (const u8 *my_png)
Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureJPG](#) (const u8 *my_jpg)
Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureJPGEx](#) (const u8 *my_jpg, const u32 my_size)
Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureBMP](#) (const u8 *my_bmp)
Load a texture from a buffer.
- [GRRLIB_texImg](#) * [GRRLIB_LoadTextureTPL](#) (const u8 *my_tpl, u32 id)
Set TPL data to a [GRRLIB_texImg](#) structure.
- bool [GRRLIB_GeckoInit](#) (void)
Initialize USB Gecko.
- void [GRRLIB_GeckoPrintf](#) (const char *text,...)
Print Gecko.
- void [GRRLIB_SetBackgroundColour](#) (u8 r, u8 g, u8 b, u8 a)
Set the background parameter when screen is cleared.
- void [GRRLIB_Camera3dSettings](#) (f32 posx, f32 posy, f32 posz, f32 upx, f32 upy, f32 upz, f32 lookx, f32 looky, f32 lookz)
Set the camera parameter (contributed by chris_c aka DaShAmAn).
- void [GRRLIB_3dMode](#) (f32 minDist, f32 maxDist, f32 fov, bool texturemode, bool normalmode)
Set up the position matrix (contributed by chris_c aka DaShAmAn).
- void [GRRLIB_2dMode](#) (void)
Go back to 2D mode (contributed by chris_c aka DaShAmAn).
- void [GRRLIB_ObjectViewBegin](#) (void)
Init the object matrix to draw object.
- void [GRRLIB_ObjectViewScale](#) (f32 scalx, f32 scaly, f32 scalz)
Scale the object matrix to draw object.
- void [GRRLIB_ObjectViewRotate](#) (f32 angx, f32 angy, f32 angz)
Rotate the object matrix to draw object.
- void [GRRLIB_ObjectViewTrans](#) (f32 posx, f32 posy, f32 posz)
Translate the object matrix to draw object.
- void [GRRLIB_ObjectViewEnd](#) (void)
Concat the object and the view matrix and calculate the inverse normal matrix.
- void [GRRLIB_ObjectView](#) (f32 posx, f32 posy, f32 posz, f32 angx, f32 angy, f32 angz, f32 scalx, f32 scaly, f32 scalz)
Set the view matrix to draw object (in this order scale, rotate AND trans).
- void [GRRLIB_ObjectViewInv](#) (f32 posx, f32 posy, f32 posz, f32 angx, f32 angy, f32 angz, f32 scalx, f32 scaly, f32 scalz)
Set the view matrix to draw object (in this order scale, trans AND rotate).
- void [GRRLIB_SetTexture](#) ([GRRLIB_texImg](#) *tex, bool rep)
Set the texture to an object (contributed by chris_c aka DaShAmAn).
- void [GRRLIB_DrawTorus](#) (f32 r, f32 R, int nsides, int rings, bool filled, u32 col)
Draw a torus (with normal).

- void [GRRLIB_DrawSphere](#) (f32 r, int lats, int longs, bool filled, u32 col)
Draw a sphere (with normal).
- void [GRRLIB_DrawCube](#) (f32 size, bool filled, u32 col)
Draw a cube (with normal).
- void [GRRLIB_DrawCylinder](#) (f32 r, f32 h, u16 d, bool filled, u32 col)
Draw a cylinder (with normal).
- void [GRRLIB_DrawCone](#) (f32 r, f32 h, u16 d, bool filled, u32 col)
Draw a cone (with normal).
- void [GRRLIB_DrawTessPanel](#) (f32 w, f32 wstep, f32 h, f32 hstep, bool filled, u32 col)
Draw a Tesselated panel (with normal).
- void [GRRLIB_SetLightAmbient](#) (u32 ambientcolor)
Set ambient color.
- void [GRRLIB_SetLightDiff](#) (u8 num, guVector pos, f32 distattn, f32 brightness, u32 lightcolor)
Set diffuse light parameters.
- void [GRRLIB_SetLightSpec](#) (u8 num, guVector dir, f32 shininess, u32 lightcolor, u32 speccolor)
Set specular light parameters.
- void [GRRLIB_SetLightSpot](#) (u8 num, guVector pos, guVector lookat, f32 angAttn0, f32 angAttn1, f32 angAttn2, f32 distAttn0, f32 distAttn1, f32 distAttn2, u32 lightcolor)
Set Spot light parameters.
- void [GRRLIB_SetLightOff](#) (void)
Set all lights off, like at init.
- [GRRLIB_ttfFont](#) * [GRRLIB_LoadTTF](#) (const u8 *file_base, s32 file_size)
Load a TTF from a buffer.
- void [GRRLIB_FreeTTF](#) ([GRRLIB_ttfFont](#) *myFont)
Free memory allocated by TTF fonts.
- void [GRRLIB_PrintfTTF](#) (int x, int y, [GRRLIB_ttfFont](#) *myFont, const char *string, unsigned int fontSize, const u32 color)
Print function for TTF font.
- void [GRRLIB_PrintfTTFW](#) (int x, int y, [GRRLIB_ttfFont](#) *myFont, const wchar_t *string, unsigned int fontSize, const u32 color)
Print function for TTF font.
- u32 [GRRLIB_WidthTTF](#) ([GRRLIB_ttfFont](#) *myFont, const char *, unsigned int)
Get the width of a text in pixel.
- u32 [GRRLIB_WidthTTFW](#) ([GRRLIB_ttfFont](#) *myFont, const wchar_t *, unsigned int)
Get the width of a text in pixel.
- void [GRRLIB_PrintfSystemFont](#) (int x, int y, const char *string, unsigned int fontSize, const u32 color)
Print text using the system font.

9.3.1 Detailed Description

GRRLIB library function prototypes.

Do not include [GRRLIB__lib.h](#) directly, include only [GRRLIB.h](#).

Chapter 10

Examples

10.1 template/source/main.c

This example shows the minimum code required to use GRRLIB.

This example shows the minimum code required to use GRRLIB. It could be used as a template to start a new project. More elaborate examples can be found inside the *examples* folder.

```
/*=====
    GRRLIB (GX Version)
    - Template Code -

    Minimum Code To Use GRRLIB
    =====*/
#include <grrlib.h>

#include <stdlib.h>
#include <wiiuse/wpad.h>

int main(int argc, char **argv) {
    // Initialise the Graphics & Video subsystem
    GRRLIB_Init();

    // Initialise the Wiimotes
    WPAD_Init();

    // Loop forever
    while(SYS_MainLoop()) {

        WPAD_ScanPads(); // Scan the Wiimotes

        // If [HOME] was pressed on the first Wiimote, break out of the loop
        if (WPAD_ButtonsDown(0) & WPAD_BUTTON_HOME) break;

        // -----
        // Place your drawing code here
        // -----

        GRRLIB_Render(); // Render the frame buffer to the TV
    }

    GRRLIB_Exit(); // Be a good boy, clear the memory allocated by GRRLIB
    exit(0); // Use exit() to exit a program, do not use 'return' from main()
}
```


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